

MME 305
Lab Report:
Pancake Project

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April 29, 2026



Executive Summary

The goals for this project include a detailed and efficient measurement system, with an easy and intuitive MATLAB app that any user can operate. The measurement system worked off using a Tare to set the load cell to 0 after the user puts the cup on it which allows for the user to get an accurate weight of each ingredient. Following the first ingredient the MATLAB code would take the difference of the previous weight with the new weight after the addition of the ingredient. The code also has a visual cue for the right amount of ingredient added making it so that the user can look at the app while they add the ingredient to check that the amount is correct, based on a red and green light. Once completing each ingredient the user would see that each ingredient had the indicator that was green showing that they were ready to move onto the next step.

Measurement System Design:

This section should explain how the sensors, transducers, and/or amplifiers are connected to the data acquisition system. Detailed information (such as wiring diagrams, calibration sheets, a bill of materials, and budget) may be referenced here but should be placed within an appendix. These appendices should be detailed enough so the reader could recreate the measurement system.

The Loadcell system makes use of the Uxcell 2kg loadcell, a linear voltage regulator (LVR), an instrumentation amplifier (inamp), and a DROK power converter as shown in Fig. 2. Firstly 5V of DC power is supplied to the DROK power converter where it is then turned into $\pm 15\text{VDC}$. The $+15\text{VDC}$ is then directed to the LVR to port 8 of the instrument amplifier, which is used to power the inamp and the -15VDC is wired directly to port 5 of the inamp, which is the negative supply voltage. The LVR converts the $+15\text{VDC}$

to +5VDC. The +5VDC is then connected with a red wire to the power supply of the 2 kg load cell and a black wire to reference ground.

A white wire and green wire leave the load cell carrying the signal generated by the weights placed on the load cell. The white wire enters port 1 on the inamp and the green enters port 4. Port 1 and 4 are the negative and positive signal inputs respectively. The gain resistor in ports 2 and 3 creates gain within the signal so that it can be easily read. The signal then leaves port 7, where it is fed through a low pass filter to further smooth out the signal, and then into the NI-DAQ where the analog signal is converted to a digital signal and read by the computer.

Similarly, the Thermocouple also uses the same DC power supply and the $\pm 15\text{VDC}$ generated by the DROK power converter. The +15VDC is connected to the adafruit board that is used in the thermocouple system. The positive and negative terminals of the adafruitboard are also connected to the type K thermocouple, which is the griddle's temperature sensor. The output of the adafruitboard is connected to a chosen analog port of the NI-DAW along with its ground as reference. Additionally, the digital outputs of the DAQ are connected to the Steady state relay (SSR) which functions as a switch, turning on the griddle.

The blender is also connected to the SSR as well as +5VDC (in order to function). The SSR serves as a switch for the blender too, activating it when called by the computer software.

Description of Work Flow:

The MATLAB App provides a guided system that allows for the pancakes to be made through a manufacturing process. It starts with the Temperature Control panel, this is where the user can turn on the griddle using a digital switch. The app will continually read the input from the thermocouple (analog), and convert the voltage into temperature, showing the value right above the digital switch. There is also a feedback control loop that keeps the griddle's temperature within the range of 374-376 °F by toggling a digital output that controls the heater. This app also stores the voltage and converted

temperature over time so that it can be exported onto an excel file for later.

The user is then guided to the Ingredient Control panel, this is where the load cell is used to measure the weights of each individual ingredient. The user is first told to tare the system after putting the blender cup on the load cell, this is to get an accurate measurement of each ingredient. Then, by selecting each ingredient from the dropdown such as milk, egg, flour, sugar, or butter the app tracks the weight against the acceptable weights (built in). After it compares the weight it will give a visual feedback by changing from a red light to a green light. During this process the app records the load cell voltage and converts it using a calibration equation, keeping these values for the excel spreadsheet.

After the ingredients are added to the blender the Blender panel can start mixing after the user activates the digital output switch. Once the switch is activated the blender runs for 15 seconds, resetting the switch automatically. The app still continues to record the weight and temperature during the blender process.

During the cooking phase the user can initiate the timers for each side of the pancake in the Timer panel. The first side runs for 3 minutes with a visual indicator (green light) prompting the user when they should flip and when the pancake is done. Also there is an image that changes to show a finished pancake to show the user it's done. The app continues to keep track of the temperature to ensure the griddle has the right temperature.

Throughout the entire process, the app collects the time-series data including analog voltage, weight, temperature, and averages as well as standard deviations. At the end the user can select the Export Data button to get the recorded data into a table and save it as an Excel file. In this file it includes a dataset of the pancake manufacturing process.

User Experience

The pancake app drastically improved both the efficiency of creating a pancake and ease of use for those unfamiliar with electronic equipment. For example, the use of our app's griddle switch in conjunction with the app timer ensured that the temperature of the griddle stayed within the correct temperature for cooking during a given time period. This makes the process of cooking identical pancakes repeatable and easy to use for someone who has never seen a thermocouple. Without this standardization, it would be easy for someone to forget to flip their pancakes, and burn them.

Additionally, our app boasts individual ingredient weighings with automatic tare of the loadcell, to ensure that the correct amount of ingredients are being used for each pancake, increasing process repeatability compared to lower accuracy of measuring cups. Again, this allows the process to be repeatable, but also allows the user to correct human error in case ingredients are weighed out of chronological order of the ingredient list, or if ingredients need to be re-measured at a later time.

Finally, our app has a blender switch that shuts off the blender automatically after the ingredients have been well blended. This allows for more consistent pancakes as it minimizes user error. Additionally it also benefits the user's ease of use, as it is one less step that they need to actively think about.

Conclusions

The results of this project show that integrating sensors and data acquisition with a MATLAB app designed for users, improves repeatability and accuracy of the pancake manufacturing process. The measurement system was able to capture real-time data for all the values needed by converting signals into accurate values through calibration. With the load cell having a tare feature it was able to get precise ingredient measurement, while the griddle maintained the 374-376 °F temperature. Additionally, the data being logged throughout the process was able to be stored and exported to an excel file.

The MATLAB app has a higher efficiency than the user relying on judgement alone. For example the taring simplifies values and setup and the blender automatically stopping after the time removes the likelihood of human error. Along with the reduction in decisions the user has to make creates a better workflow. The app also has visual cues and real-time feedback, such as the red light and green light, allowing the user to verify the amount they add.

The app standardizes each step, and the ingredient amounts are able to be checked against the predefined range. Temperature is actively regulated and is able to be monitored via the app. Cooking times are fixed, removing variability caused by human estimation, making these pancakes gnarly identical to each other over multiple tests. This project demonstrates that this system can be controlled and repeatable which in turn improves both consistency and user experience.

Appendix

Item #	Part Name	Manufacturer	Quantity	Unit Cost (\$)	Total Cost (\$)
1	BreadBoard	Generic (various)	1	\$15	
2	Jumper Wires Bundle	Generic (Elegoo / generic kits)	1	\$12	
3	Terminal Blocks	Generic (Phoenix Contact / clones)	12	\$60	
4	Male Barrel Jack Connectors	CUI Devices / Switchcraft / generic	2	\$8	
5	Female Barrel Jack Connectors	CUI Devices / Switchcraft / generic	3	\$12	
6	LM7805CT Linear Voltage Regulator	Texas Instruments / ON Semiconductor / STMicroelectronics	1	\$2	
7	Solid-State Relay	Omron / Crydom (Sensata)	1	\$25	
8	DROK +/- 15V Power Converter	DROK	1	\$25	
9	5V 2A Power Supply	Generic (Mean Well / generic)	1	\$8	
10	K-Type Thermocouple	Omega Engineering / generic	1	\$10	
11	Uxcell 2 kg Load Cell	Uxcell	1	\$15	
12	K-Type Thermocouple Amplifier	Adafruit / Analog Devices-based modules	1	\$10	
13	Adafruit 1778 Board	Adafruit	1	\$16	
14	AD 8429 Instrumentation Amplifier	Analog Devices	1	\$10	
15	NI USB 60001	National Instruments	1	\$150	
16	USB Cable	Generic (Anker / Amazon Basics / Belkin)	1	\$2	
17	MATLAB and NI MAX	MathWorks (MATLAB) / National Instruments (NI MAX)	1	\$100	\$480

Fig 1. Bill of Materials

MME305 Measurements & Instrumentation Calibration Sheet
Pancake Project, Load / Torque Cells

LOAD CELL CALIBRATION SHEET

Instrumentation

Item	Details
Instrument Type	Load Cell
Rated Capacity	2 kg
Manufacturer	Uxcell
Model	1034748
Serial Number	D1
Calibration Date	04/08/26

Location	GAR 145, Miami University, 650 E High St. Oxford Ohio
Environmental Conditions	Temp: RT
DAQ Used	NI USB 6001
Excitation Voltage	5V

Weights Used

Reference Weights	1	2	3	4	5
Weight (g)	0	20	40	100	200

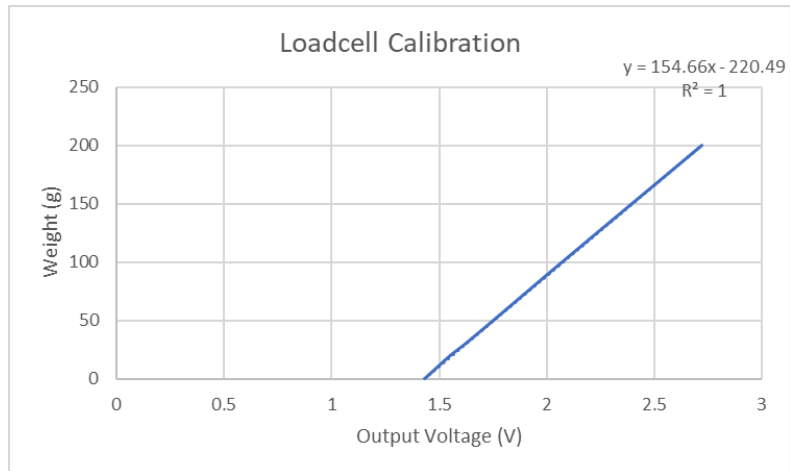
Calibration Method

Method: Weights of 5 known masses were placed one at a time upon the center of the load cell in no particular order. The voltage output was read on the DAQ for each weight and then recorded in an excel document. Using this data, a plot was constructed and trendline was formed to give a calibration constant (the slope of the curve) and a zero load offset which was displayed as the y intercept.

Calibration Results

Parameter	Tolerance	Result
Calibration Constant	NA	154g/V
Zero Drift	$\pm 0.05\%FS$	-220.5V

Calibration Curve



Signatures

Technician Signature: Kyle Buttray Date: 4/08/26

Technician Signature: Troy Jones Date: 4/08/26

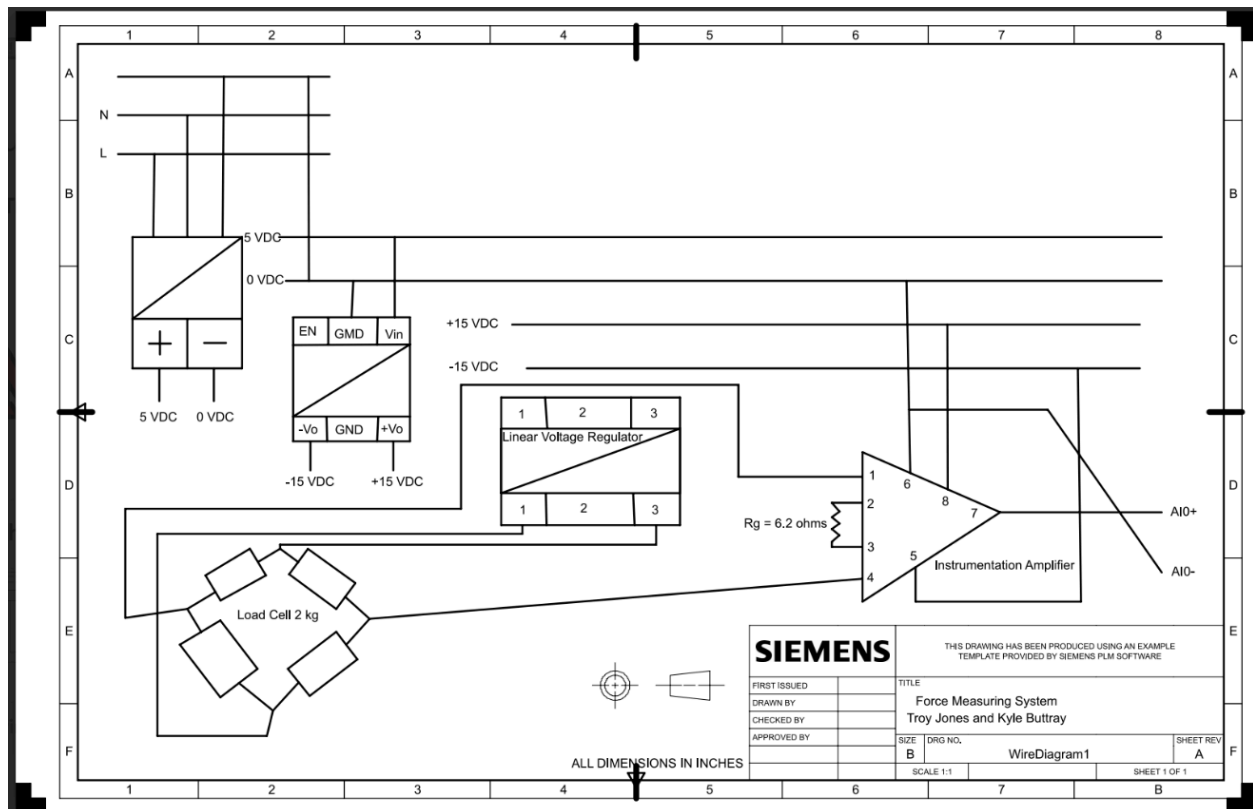


Fig 2. Loadcell Wiring Diagram

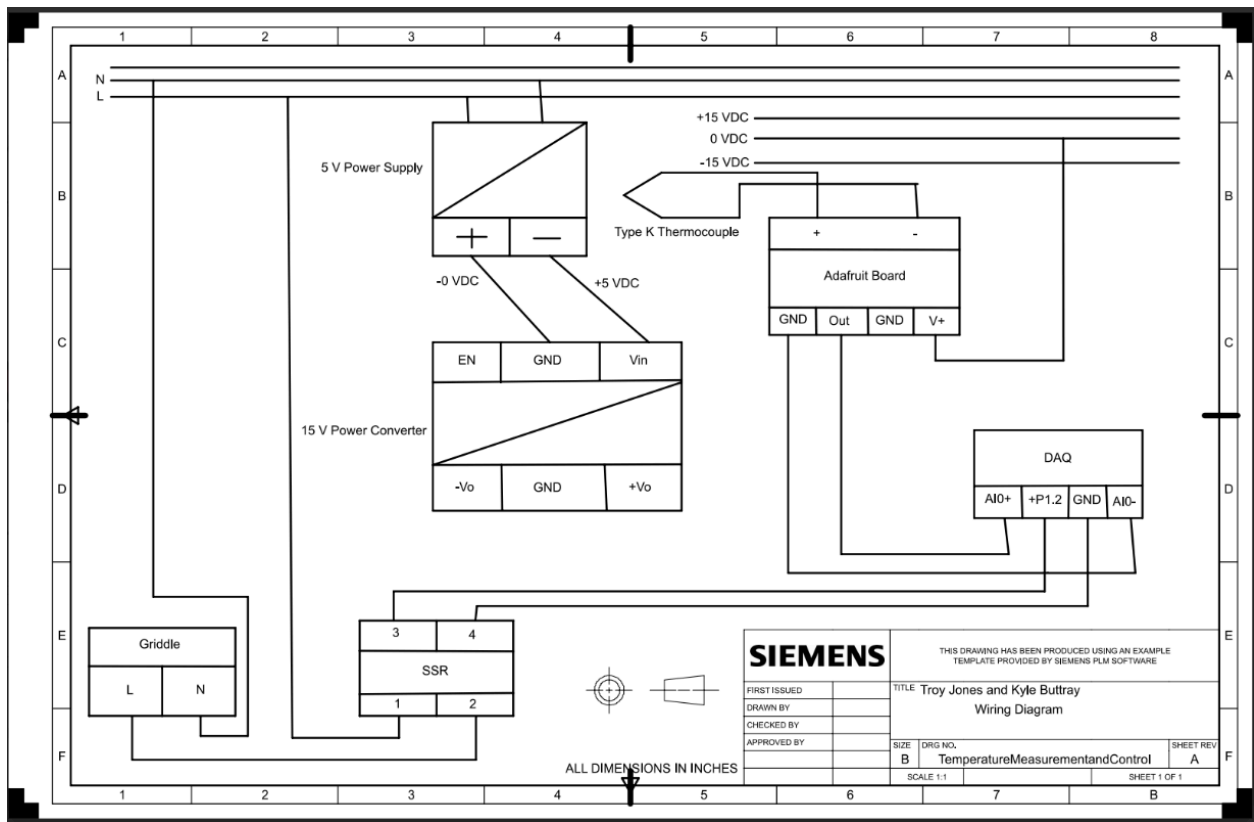


Fig 3. Thermocouple Wiring Diagram

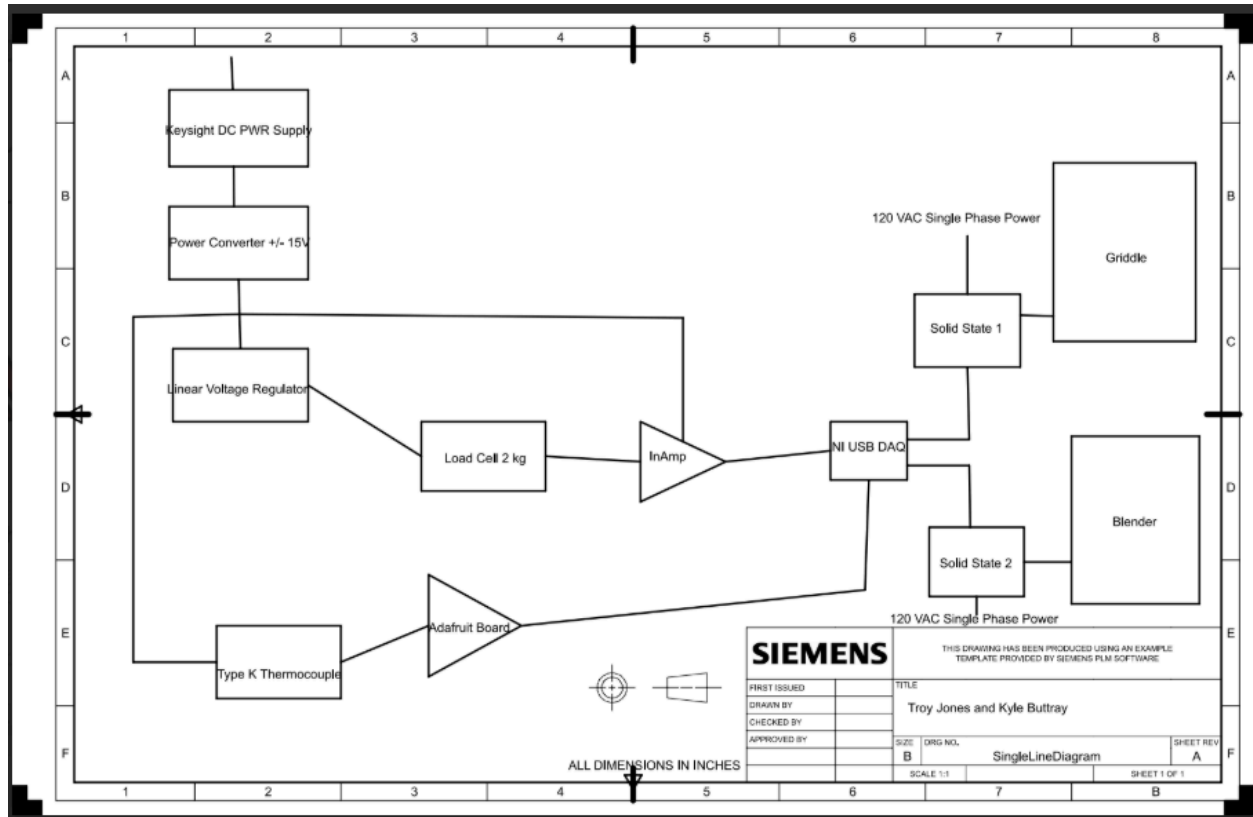


Fig 4. Single Line Diagram

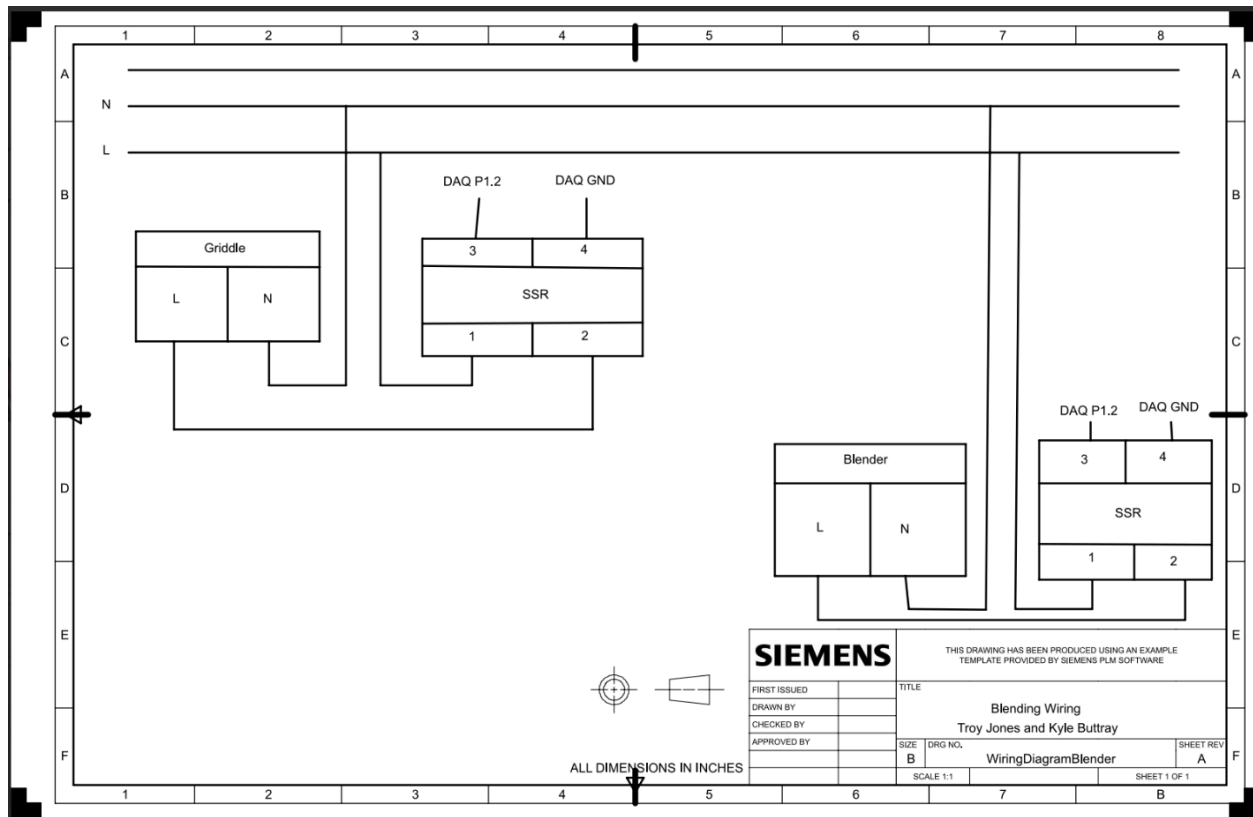


Fig 5. Blender and Griddle Wiring Diagram

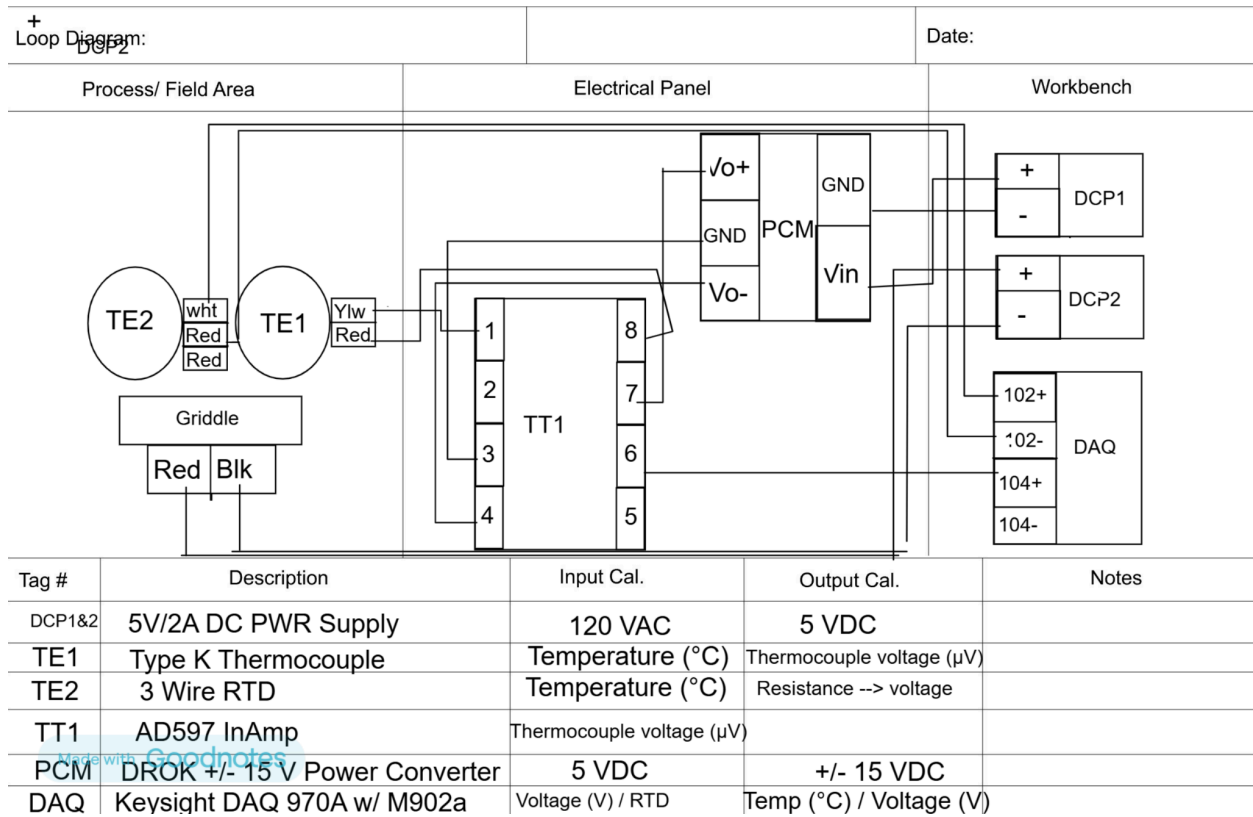


Fig. 6 Thermocouple Loop Diagram

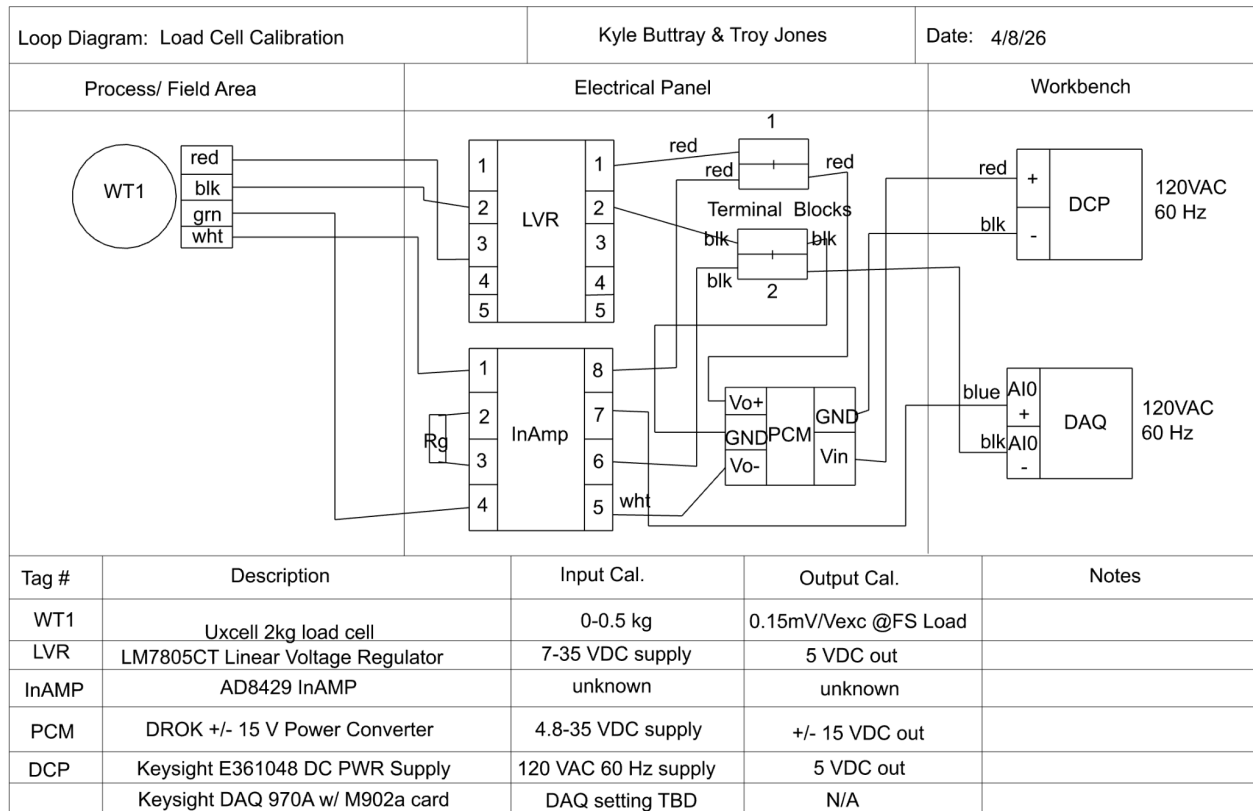


Fig. 7 Loadcell Loop Diagram